# Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

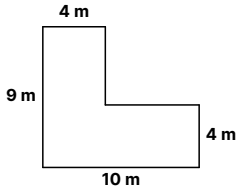
**KEY WORDS** compound shape · perimeter *(paenga)* · area *(horahanga)* · semicircle *(haurua porowhita)*

## A Shapes made from rectangles ●○○

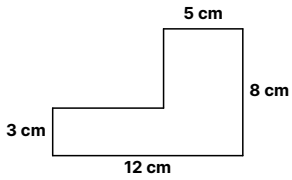
**Remember:** Find the missing side lengths first. For the area, split the shape into rectangles.

**1.** Find the perimeter and the area of each shape. All corners are right angles.

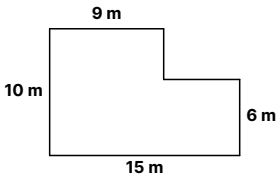
**a)**



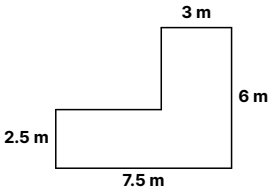

**b)**



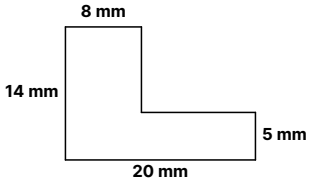

**c)**



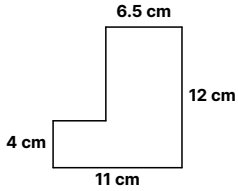

**d)**




**e)**




**f)**

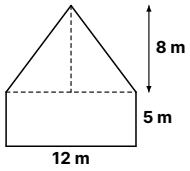



## B Shapes with triangles ●●○

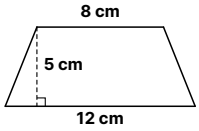
**Remember:** Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a+b) \times h$.

**2.** Find the area of each shape.

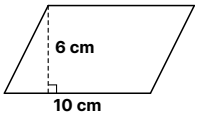
**a)**



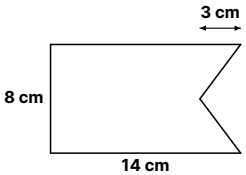

**b)**



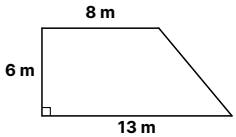

**c)**



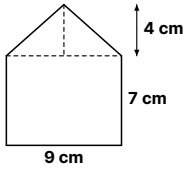

**d)**




**e)**




**f)**





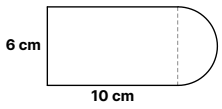

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## C Shapes with circles ●●●

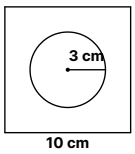
**Remember:** Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

**3.** Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

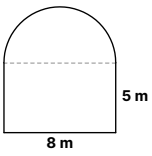
**a)**



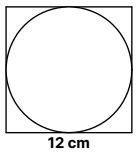

**b)**




**c)**




**d)**

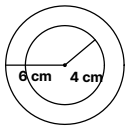



## D Shaded regions ●●●

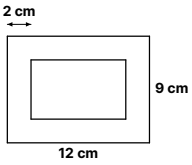
**Remember:** A shaded region is the big shape minus the hole. Find each area separately, then subtract.

**4.** Find each shaded area, to 1 decimal place where needed.

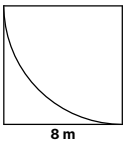
**a)**



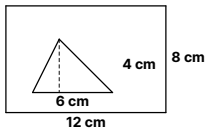

**b)**




**c)**




**d)**




## E Problems ●●●

**5.** A rectangular lawn is 12 m long and 8 m wide. A circular flower bed with a radius of 2 m is cut out of it.

**a)** Find the area of the grass, to 1 decimal place.

**b)** Fertiliser for the grass costs $3.50 for each square metre. Find the cost of fertilising the lawn.

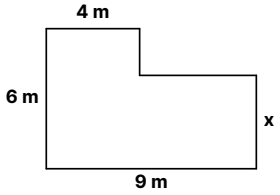
**6.** A wall is 6 m long and 2.5 m high. It has a window 1.2 m by 1.5 m and a door 0.8 m by 2 m. Find the area of wall to be painted.

**7.** A running track has two straight sides, each 100 m long, joined by a semicircle at each end. The semicircles have a diameter of 60 m. Find the distance once around the track, to the nearest metre.

## ★ Extension

**8.** A square has sides of 10 cm. A quarter circle with a radius of 5 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

**9.** The L-shape has an area of 44 m². Find the length marked $x$.






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# Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 2

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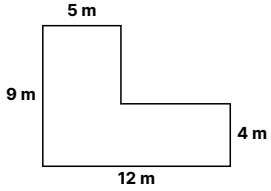
**KEY WORDS** compound shape · perimeter *(paenga)* · area *(horahanga)* · semicircle *(haurua porowhita)*

## A Shapes made from rectangles ●○○

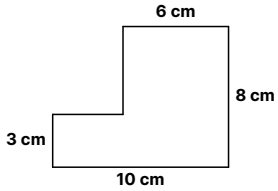
**Remember:** Find the missing side lengths first. For the area, split the shape into rectangles.

**1.** Find the perimeter and the area of each shape. All corners are right angles.

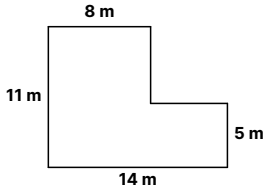
**a)**



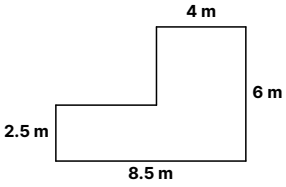

**b)**



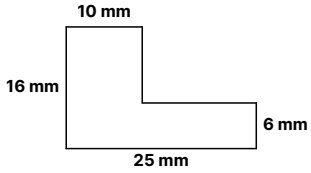

**c)**



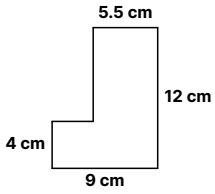

**d)**




**e)**




**f)**

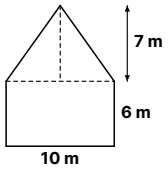



## B Shapes with triangles ●●○

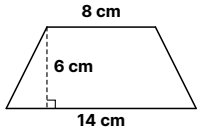
**Remember:** Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a+b) \times h$.

**2.** Find the area of each shape.

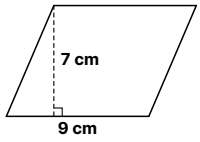
**a)**



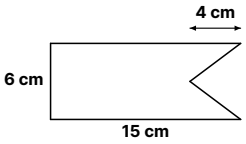

**b)**



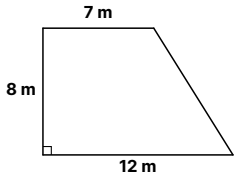

**c)**



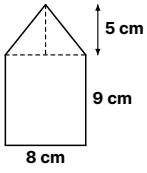

**d)**




**e)**




**f)**





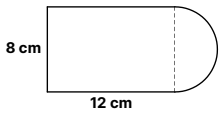

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## C Shapes with circles ●●●

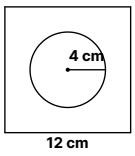
**Remember:** Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

**3.** Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

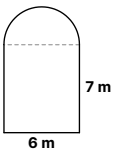
**a)**



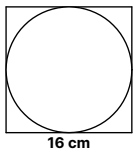

**b)**




**c)**




**d)**

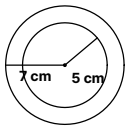



## D Shaded regions ●●●

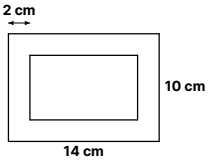
**Remember:** A shaded region is the big shape minus the hole. Find each area separately, then subtract.

**4.** Find each shaded area, to 1 decimal place where needed.

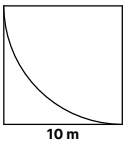
**a)**



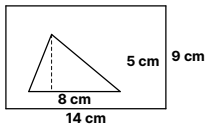

**b)**




**c)**




**d)**




## E Problems ●●●

**5.** A rectangular lawn is 15 m long and 9 m wide. A circular flower bed with a radius of 3 m is cut out of it.

**a)** Find the area of the grass, to 1 decimal place.

**b)** Fertiliser for the grass costs $4.50 for each square metre. Find the cost of fertilising the lawn.

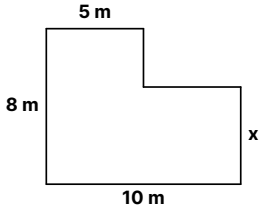
**6.** A wall is 7 m long and 2.4 m high. It has a window 1.5 m by 1.2 m and a door 0.9 m by 2 m. Find the area of wall to be painted.

**7.** A running track has two straight sides, each 80 m long, joined by a semicircle at each end. The semicircles have a diameter of 50 m. Find the distance once around the track, to the nearest metre.

## ★ Extension

**8.** A square has sides of 12 cm. A quarter circle with a radius of 6 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

**9.** The L-shape has an area of 65 m². Find the length marked $x$.






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